

Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT305	Jimin XIAO	Intelligent Science	3209

Final Exam 2022/2023

BACHELOR DEGREE – Year 4

Machine Learning

TIME ALLOWED: 3 Hours 0 Minutes

INSTRUCTIONS TO CANDIDATES

- 1、 Total marks available are 100.
- 2、 Answer all questions.
- 3、 The number in the column on the right indicates the approximate marks for each section.
- 4、 In general, it is particularly important to give reasons for your answer. Only partial marks will be awarded for correct answers with inadequate reasons.
- 5、 Answer should be written in the answer booklet(s) provided.
- 6、 The university approved calculator - Casio FS82ES/83ES can be used.
- 7、 Only solutions written in English will be accepted.

Answer All Questions

1. Pooling units take n values x_i , $i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

(1) The Lp-pooling module takes positive inputs and computes $y = (\sum_i x_i^p)^{\frac{1}{p}}$, assuming we know that

$$y' = \frac{\partial L}{\partial y}, \text{ what is } x'_i = \frac{\partial L}{\partial x_i} ?$$

(2) The log-average module computes $y = \frac{1}{\beta} \ln(\frac{1}{n} \sum_i \exp(\beta x_i))$, assuming we know that $y' = \frac{\partial L}{\partial y}$, what

$$\text{is } x'_i = \frac{\partial L}{\partial x_i} ?$$

Total

16 (each 8)

2. Recall two linear classification methods we considered:

Model 1:

$$y = \mathbf{w}^T \mathbf{x} + b$$

$$\mathcal{L}_{SE}(y, t) = \frac{1}{2} (y - t)^2$$

Model 2:

$$z = \mathbf{w}^T \mathbf{x} + b$$

$$y = \sigma(z)$$

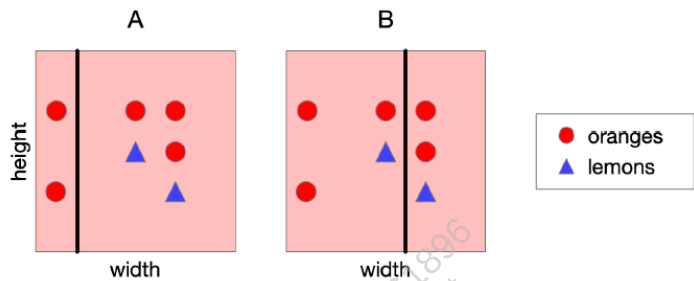
$$\mathcal{L}_{SE}(y, t) = \frac{1}{2} (y - t)^2$$

Here, σ denotes the logistic function (sigmoid function), and the target label t take values in $\{0, 1\}$. Briefly explain our reason for preferring Model 2 to Model 1.

Total

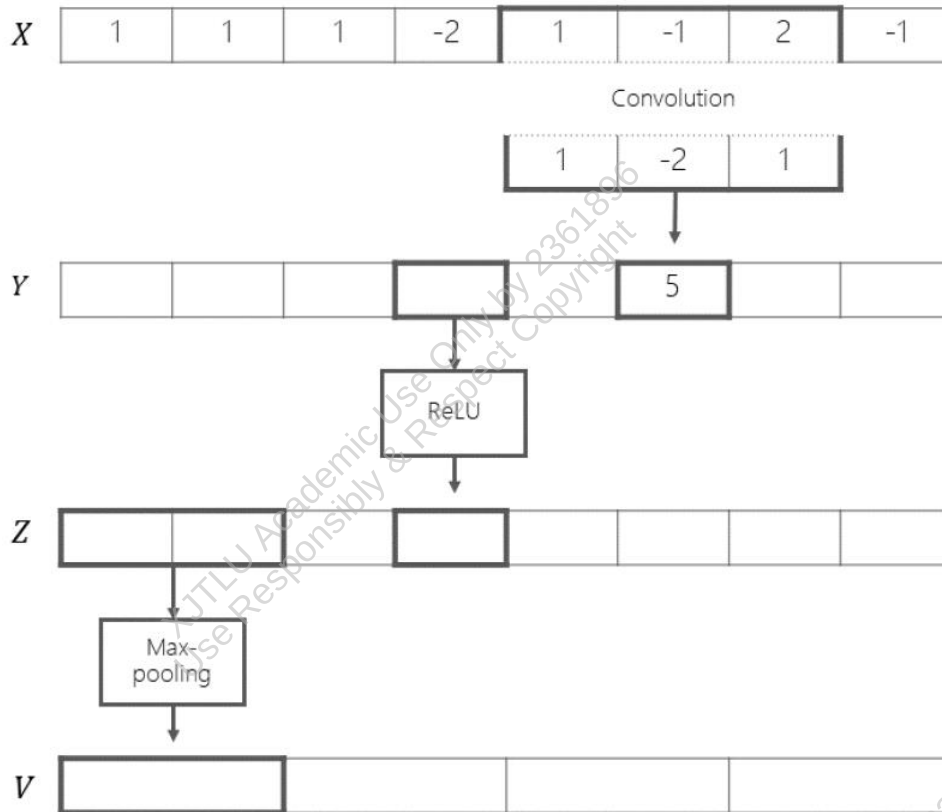
14

3. We are constructing a decision tree. Considering the following data, we would like to split them on width for one level. There are 2 options to split them as in the following figure. Please select the better one from the perspective of information gain.



Total
15

4. The sub-system shown in the following figure has a convolution unit (the convolution kernel is 1×3), followed by a ReLU, and then followed by max-pooling unit. In the following, we assume that the convolution operation uses zero padding.



Sub-system with convolution, ReLU, then followed by Max-Pooling unit.

- (1) Fill the empty elements (i.e., all the empty elements) in the vector Y .
- (2) Fill the empty elements (i.e., all the empty elements) in the vector Z .
- (3) Fill the empty elements (i.e., all the empty elements) in the vector V .
- (4) How many learnable parameters does this sub-system have?
- (5) Given the pictorial representation shown in the above figure, what is the stride size of the convolution unit?

Total

20 (each 4)

5. You want to train the following model using gradient descent. Here, the input x and target t are both scalar-valued.

$$z = \omega_0 + \omega_1 x + \omega_2 x^2 + \omega_3 x^3$$

$$y = 1 + e^{\max(z, 0)}$$

$$L = \frac{1}{2}(\ln y - \ln t)^2$$

Determine the back propagation rules which will let you compute the loss derivative of $\bar{\omega}_2 = \frac{\partial L}{\partial \omega_2}$ (the variables with the bar are the gradient of the variables). Your equations should refer to previously computed values (e.g. your formula for $\bar{z}$ should be a function of $\bar{y}$). You do not need to show your derivation. The dummy step has been filled in for you.

$$\bar{\mathcal{L}} = 1$$

$$\bar{y} =$$

$$\bar{z} =$$

$$\bar{w}_2 =$$

Total
15 (each 5)

6. In this question, you will derive the E-M update rules for a univariate Gaussian mixture model (GMM) with two mixture components. Unlike the GMMs we covered in the course, the mean μ will be shared between the two mixture components, but each component will have its own standard deviation σ_k (σ_0 and σ_1). The mixture component is indicated by a latent variable $z \in \{0,1\}$. The model is defined as follows:

$$z \sim \text{Bernoulli}(\theta) \text{ (i.e., } P(z = 1|\theta) = \theta, P(z = 0|\theta) = 1 - \theta)$$

$$x | z = k \sim \mathcal{N}(\mu, \sigma_k) \text{ for } k \in \{0,1\}$$

The parameters of the model are θ , μ , σ_0 , and σ_1 . Suppose we observe a dataset $\{x^{(i)}\}_{i=1}^N$.

For reference, the PDF of the Gaussian distribution is as follows:

$$\mathcal{N}(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- (1) Write the **complete** data log-likelihood for this model. (In this case, we assume that $z^{(i)}$ is known for every $x^{(i)}$, thus our dataset is $\{z^{(i)}, x^{(i)}\}_{i=1}^N$)
- (2) In the E-step, for each data point $x^{(i)}$, we need to compute the posterior probability $r^{(i)} = \text{Pr}(z^{(i)} = 1 | x^{(i)})$. Please Give the formula for $r^{(i)}$. In your formula, you may use $\mathcal{N}(x^{(i)}; \mu, \sigma)$ to denote the Gaussian PDF, rather than writing it out explicitly.

$$r^{(i)} =$$

- (3) Write out the expected complete data log-likelihood, i.e., the objective that is to be maximized in the M-step. It should be expressed in terms of the $r^{(i)}$ and the Gaussian PDF $\mathcal{N}(x^{(i)}; \mu, \sigma)$.
- (4) Derive the M-step update rule for μ by maximizing this objective with respect to μ . (In this step, the σ_k are fixed to their previous values.)

Total
20 (each 5)

The end of the paper